

Topic: Cloud Security Issues and

Challenges: A Survey

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Report by: Group 3 (29 - 37)

Team Members

Roll Number	Name	Signature
29	Atharv Nandagave	
30	Yash Nayak	
31	Neev Jain	
32	Ashutosh Pandey	
33	Harsh Pandey	
34	Pratyush Pandey	
36	Pancham Patil	
37	Eshaan Pilar	

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1. INTRODUCTION

Cloud Computing has emerged as a transformative paradigm that delivers on-demand computing services over the Internet, eliminating the need for users to invest in expensive physical infrastructure. The foundational concept—*computing as a utility*—was first introduced by Corbato and Vyssotsky in 1965. Today, cloud computing integrates virtualization, distributed systems, and web technologies to offer a pay-as-you-go model for resources such as CPU, memory, storage, and bandwidth.

The **National Institute of Standards and Technology (NIST)** defines cloud computing through five essential characteristics: **on-demand self-service**, **broad network access**, **resource pooling**, **rapid elasticity**, and **measured service**. These characteristics support three primary service delivery models, Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS), deployed across public, private, hybrid, and community cloud environments.

Despite its economic and operational advantages, the migration from local computing to remote, virtualized infrastructure introduces profound **security risks**. A 2009 IDC survey revealed that **87.5% of Chief Information Officers (CIOs)** regard security as the top priority inhibiting cloud adoption. Key concerns include:

- **Loss of control** over data stored in geographically distributed, opaque data centers.
- **Multi-tenancy vulnerabilities**, where shared physical resources enable co-residence and side-channel attacks.
- **Virtual Machine (VM) image vulnerabilities**, including malicious code injection and improper sanitization.
- **Insecure interfaces and APIs**, which expose cloud services to injection, spoofing, and session hijacking.
- **Insider threats** from privileged administrators within the Cloud Service Provider (CSP) organization.

Conceptual Overview of Common Cloud Attack Vectors and Entry Points

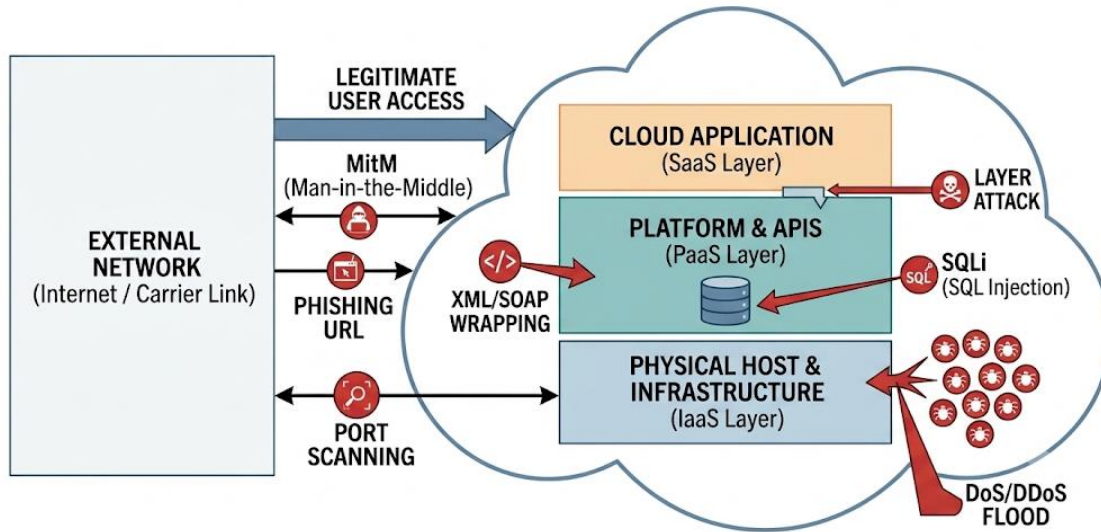


Figure 1.1: System diagram illustrating attack vectors (MitM, Phishing, SQLi, XML Wrapping, DoS/DDoS) across SaaS, PaaS, and IaaS cloud layers.

2. THEORETICAL BACKGROUND

Cloud computing is a distributed computing model that merges large numbers of heterogeneous resources into a unified system. Unlike traditional cluster or grid computing, cloud infrastructure leverages complete virtualization to abstract hardware details from the end user. The architectural framework comprises:

- **Cloud Consumer:** End users accessing services via web browsers, mobile apps, or thin clients.
- **Cloud Provider:** Organizations that own and manage cloud infrastructure (e.g., AWS, Google Cloud, Azure).
- **Cloud Broker:** An intermediary that negotiates relationships between consumers and providers.

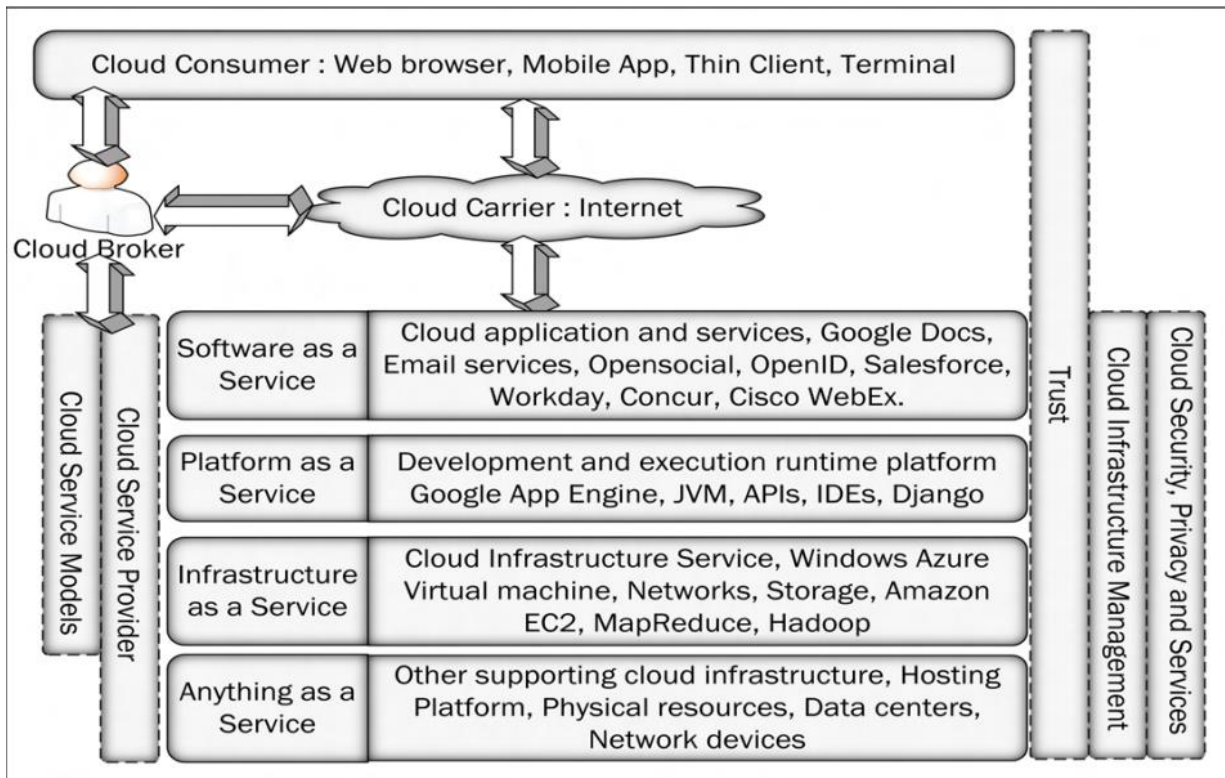


Figure 2.1: A Complete Cloud Computing Architectural Framework Shows the layered interaction between consumers, carriers, and providers across SaaS, PaaS, IaaS, and XaaS layers, bounded by trust and security management.

The NIST model defines five essential characteristics that distinguish cloud computing:

1. **On-Demand Self-Service:** Consumers provision resources automatically without human interaction.
2. **Broad Network Access:** Services are available over the network through standard mechanisms and heterogeneous client platforms.
3. **Resource Pooling:** Physical and virtual resources are dynamically assigned and reassigned across a multi-tenant environment.
4. **Rapid Elasticity:** Resources can be elastically provisioned and released to scale rapidly outward and inward with demand.

Service Delivery Models:

Model	Description	Examples
SaaS	Consumer uses provider's applications running on cloud infrastructure.	Salesforce, Google Docs, Cisco WebEx
PaaS	Consumer deploys applications onto a programmable platform managed by the provider.	Google App Engine, JVM, Django
IaaS	Consumer provisions processing, storage, networks, and other fundamental computing resources.	Amazon EC2, Windows Azure, Hadoop

Deployment Models:

- **Private Cloud:** Operated exclusively for a single organization.
- **Public Cloud:** Owned and managed by a CSP; resources are shared among the general public.
- **Community Cloud:** Shared among several organizations with common concerns.
- **Hybrid Cloud:** Composition of two or more distinct cloud infrastructures bound by standardized technology.

Data in the cloud is stored through three primary mechanisms:

- **Shared File / Block Storage:** Uses protocols such as NFS, SMB, and CIFS for multi-user file access.
- **Object Storage:** Data is stored as objects accessed via global keys, hashes, or URLs using REST/HTTP protocols (e.g., CDMI).
- **Database / Table Storage:** Relational databases (RDBMS) maintaining data integrity through row-column structures; scaling remains a challenge..

Top Threats (CSA 2013):

Threat	Description	Affected Models
Abuse & Nefarious Use	Fraudulent registration to launch DDoS, password cracking, or host malicious data.	IaaS, PaaS
Insecure Interfaces / APIs	Weak authentication or improper access control over management interfaces.	SaaS, PaaS, IaaS
Malicious Insiders	Insider abuse of elevated privileges leading to data breaches.	All models
Shared Technology Issues	Hypervisor or infrastructure compromise affecting multiple tenants.	IaaS
Data Loss / Leakage	Deletion, alteration, or theft without adequate backup or encryption.	All models
Account/Service Hijacking	Credential theft via phishing or exploitation, leading to session compromise.	All models

Common Attack Vectors:

- **DoS / DDoS:** SYN floods, UDP floods, Smurf attacks, and resource exhaustion via XML flooding.
- **Service Injection:** Insertion of malicious VMs or code into the service provisioning pipeline.
- **VM Escape / Hypervisor Attack:** Gaining control of the VMM to monitor or shut down other VMs.
- **Man-in-the-Middle (MitM):** Interception of insecure SSL/TLS channels.
- **Metadata Spoofing:** Modification of WSDL or service descriptor files to redirect legitimate requests.
- **Phishing & Backdoor Channels:** Social engineering and persistent remote access tools.

3. LITERATURE SURVEY

Evolution of Cloud Security Research

Early distributed systems relied on clusters and grids, which were expensive or management intensive. Cloud computing (Circa 2008) introduced utility-based resource delivery but inherited and amplified security challenges from its predecessors. Prior to 2010, research focused primarily on grid security and virtualization isolation. Post-2010, the proliferation of public clouds shifted academic focus toward multi-tenancy risks, API security, and data privacy in outsourced environments.

Comparative Analysis of Existing Surveys

Numerous surveys have addressed cloud security; however, most focus on isolated domains such as virtualization, trust management, or specific service models. Table 3.1 presents a comparative summary of prominent survey papers and their coverage across critical dimensions.

Survey paper	Year	Topic discussed	Cloud overview	Cloud automation	Security requirements	Cloud trust	Cloud security (abstraction, threats & attacks)	Cloud security issues	Security solutions	Open issues
Takabi et al. [159]	2010	Virtualization, trust management, secure service management, security & privacy	-	X	X	✓	-	-	X	X
Zhou et al. [196]	2010	Security requirements (availability, confidentiality, integrity, control, and audit), privacy	-	X	✓	X	X	X	X	X
Grobauer et al. [59]	2011	Cloud specific vulnerability, cloud risk, authentication, authorization, and access control	-	-	-	X	X	-	X	X
Vaquero et al. [171]	2011	IaaS cloud security, cloud threats and attacks, VMM security	X	X	X	-	✓	-	-	X
Behl [24]	2011	Cloud security issues, cloud security challenges	-	X	X	X	X	-	X	✓
Zissis et al. [198]	2012	Cloud trust, cryptographic method for security, security requirements, trusted third party authentication, certification based authorization	-	X	✓	✓	-	X	X	X
Modi et al. [107]	2013	Cloud threats and attacks, security issues at different layer, authentication and virtualization security issues	X	X	✓	✓	✓	-	-	✓
Oktay et al. [115]	2013	Cloud security attacks, Intrusion detection system, intrusion prevention system	✓	X	X	X	✓	X	X	X
Fernandes et al. [49]	2014	Cloud concept and technologies, cloud security issues	✓	✓	✓	✓	X	✓	X	✓
Abbas et al. [1]	2014	Cloud security requirements, privacy preserving approaches, open research issues in E-health cloud	X	X	✓	X	X	X	X	✓
Ali et al. [6]	2015	Cloud computing overview, cloud security issues, cloud security solutions, security issues in mobile cloud computing	✓	X	X	-	-	-	-	✓
Tari et al. [163]	2015	Security issues present in public and private clouds, service availability, access control, multi-tenant services, identity and data protection	X	X	X	X	-	-	-	X
This survey		Cloud overview, cloud technologies, cloud security requirements, cloud trust, Cloud security (abstraction, threats & attacks), cloud security issues and their solutions, future research directions	✓	✓	✓	✓	✓	✓	✓	✓

Table 3.1: Comparative Analysis of Related Survey

Symbols: “✓” = Covered; “X” = Not Covered; “-” = Limited Discussion.

Key Observations:

- * Most pre-2012 surveys lack comprehensive coverage of **security solutions** alongside **threats and attacks**.
- * **Open issues** and **cloud automation** are frequently omitted.
- * The surveyed paper bridges these gaps by providing an integrated view of cloud architecture, technologies, threats, attacks, issues, solutions, and future research directions.

Research Gap and Contribution

While existing literature successfully identifies individual security domains, a unified treatment linking **cloud-specific vulnerabilities** to **actionable countermeasures** remains sparse. Furthermore, the dynamic interplay between virtualization, multi-tenancy, and legal compliance is underrepresented. This survey addresses the gap by systematically categorizing eight major security issue domains and mapping each to practical solutions and open challenges.

4. METHODOLOGY

The methodology adopted in this survey explains the systematic classification, analysis, and synthesis of cloud security literature. The approach integrates architectural decomposition, threat modeling, and comparative evaluation to produce a holistic security framework.

Research Approach and Classification Framework

The survey employs a **three-phase methodology**:

1. **Literature Collection & Filtering:** Peer-reviewed journals, conference proceedings, and industry reports (2009–2015) were analyzed. Papers were filtered based on relevance to cloud security architecture, service models, and deployment scenarios.
2. **Taxonomic Classification:** Security issues were organized into eight principal categories derived from the NIST cloud model and CSA threat reports.

Hierarchical Summary of Security Issues:

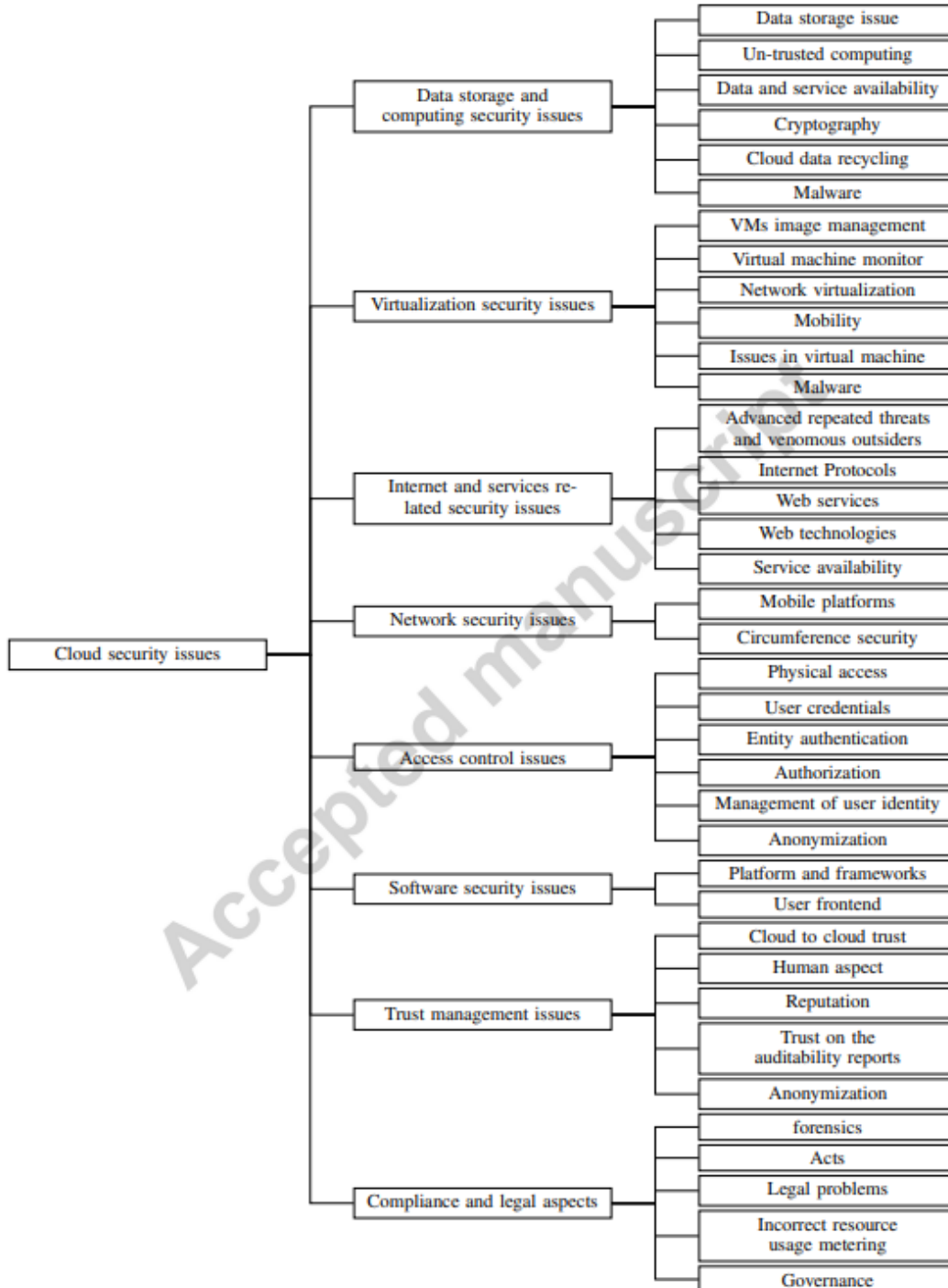


Figure 4.1: Hierarchical Classification of Cloud Security Issues Depicts the eight primary branches: Data Storage & Computing, Virtualization, Internet & Services, Network, Access Control, Software, Trust Management, and Compliance/Legal.

Virtualization Security Issues

Virtualization is the primary enabler of cloud scalability, yet it introduces unique attack vectors.

Security Topic	Key Issues	Solutions
VM Image Management	Malicious image injection, VM sprawl, repository oversight	Cryptographic image verification, privacy-preserving repositories
VMM / Hypervisor	VM escape (BLUEPILL, SubVirt), single point of failure, zero-day vulnerabilities	HyperCheck, DeHype, HyperLock, NoHype, SplitVisor
Network Virtualization	Promiscuous mode, packet sniffing, VLAN security degradation	Cisco Virtual Security Gateway, tree-rule firewalls
Mobility	Live migration interception, replay attacks, configuration drift	vTPM-based migration protocols, secure live migration frameworks
Issues in VM	Cross-VM side channels, covert channels, memory deduplication attacks	CloudVisor, CloudSec, secure runtime environments

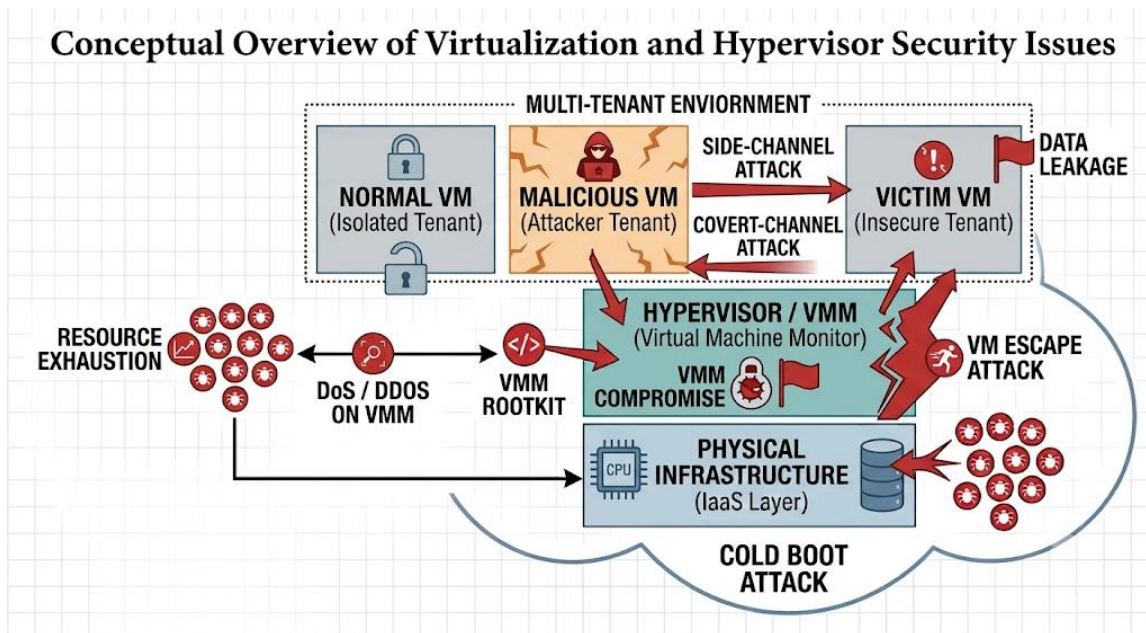


Figure 5.1 Mapping virtualization and hypervisor threats, including side-channel attacks, VM escape, hypervisor compromise, and physical infrastructure exploits.

Access Control Issues

Security Topic	Key Issues	Solutions
Physical Access	Malicious insiders, cold boot attacks, hardware tampering	XACML policies, secure facility design
User Credentials	LDAP/AD management overhead, weak reset methods	Attribute-based encryption, hierarchical access control
Authentication	Brute-force, dictionary attacks, archaic static passwords	Multi-factor authentication, SMS recovery, SSO/SAML

5. FUTURE SCOPE

While this survey provides a comprehensive taxonomy of cloud security issues, several domains require further investigation to keep pace with evolving threats and technologies.

- Integrated and Extensible Security Solutions

Current research often addresses individual issues in isolation. Future work must focus on **unified security frameworks** that simultaneously address confidentiality, integrity, availability, and auditability across all service layers without introducing operational conflicts.

- Multi-Tenancy Privacy and Isolation

As multi-tenancy remains central to cloud economics, ensuring robust **logical and virtual isolation** is critical. Future research should explore hardware-assisted isolation (e.g., Intel SGX, ARM TrustZone) and advanced side-channel mitigation techniques.

- Privacy-Preserving Computation

Most cloud operations require plaintext data for processing. **Homomorphic encryption** and **secure multi-party computation** are promising but computationally expensive. Developing practical, high-performance privacy-preserving computation models is a key open challenge.

- Standardization and Inter-Cloud Migration

The lack of standardized protocols for data and application migration fosters vendor lock-in. Future efforts must establish **open standards for VM portability**, data format interoperability, and cross-cloud identity federation.

- Insider Threat Detection and User Profiling

Distinguishing between legitimate users and malicious insiders in a shared, virtualized environment remains difficult. Research into **behavioral analytics**, **runtime assurance mechanisms**, and **automated anomaly detection** is essential.

- Legal Frameworks and Runtime SLA Assurance

Current SLAs lack standardized security metrics and enforcement mechanisms. Future work should develop **machine-auditable SLAs**, real-time compliance monitoring, and international treaties governing cross-border data jurisdiction.

6. CONCLUSION

Cloud computing offers undeniable benefits—including rapid deployment, cost efficiency, elastic scalability, and ubiquitous access. However, the shift from localized, physically controlled infrastructure to remote, virtualized, and multi-tenant environments introduces a complex landscape of security challenges.

This report has presented a comprehensive study of cloud security issues derived from a systematic survey of the literature. The theoretical background established the architectural, technological, and conceptual foundations of cloud computing. The literature survey highlighted gaps in prior work, demonstrating the need for an integrated treatment of threats, attacks, and solutions. The methodology provided a hierarchical classification of eight major security domains—ranging from data storage and virtualization to trust management and legal compliance—each mapped to concrete countermeasures.

Key findings confirm that **multi-tenancy** and **virtualization** are double-edged swords: they enable resource efficiency but amplify risks such as cross-VM side channels, hypervisor escapes, and co-residence attacks. Furthermore, **human factors**, **insecure APIs**, and **ambiguous SLAs** remain persistent vulnerabilities that technology alone cannot resolve.

The future scope outlined in this report points toward integrated security architectures, hardware-assisted isolation, privacy-preserving computation, and standardized legal frameworks. Addressing these open issues will be crucial for fostering trust and accelerating the adoption of cloud computing in security-sensitive industries such as healthcare, finance, and national research infrastructure.

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